

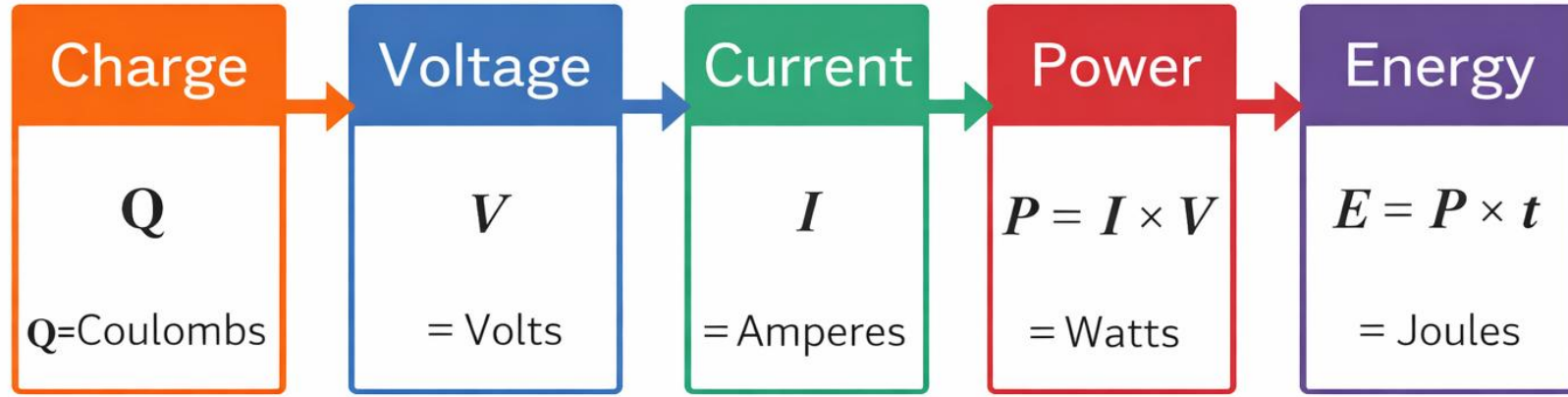
ES -116

Principles and Applications of Electrical Engineering

Semester II, 2025-26

Fundamental Laws of Electrical Circuit Analysis

Recap (Electrical Lingo)

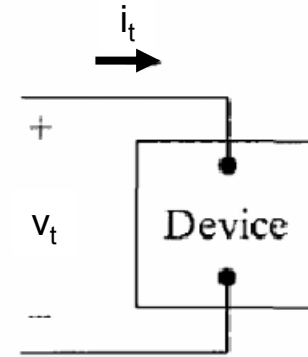
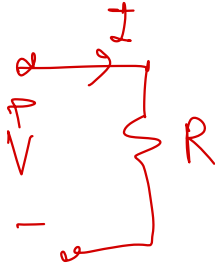


Q. A 2-wheeler EV uses a battery whose terminal voltage drops linearly from 48 V to 36 V over a 30-minute ride, while the motor draws a constant 20 A during this interval.

- Write $V(t)$ in this interval (take t in seconds, $t=0$ at 48 V).
- Express the instantaneous power $P(t)$.
- Find the total electrical energy delivered by the battery in these 30 minutes.

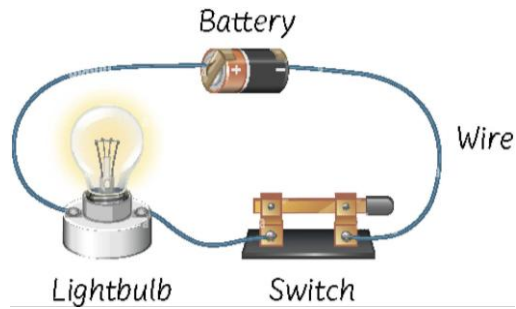
Question

The voltage and current are measured at the terminals of the device illustrated in the figure and the values of v_t and i_t are tabulated. Construct a circuit model of the device inside the box.



v_t (V)	i_t (A)
-40	-10
-20	-5
0	0
20	5
40	10

Calculating quantities in combination of bulbs



LED Spring Lights, commonly used during Diwali

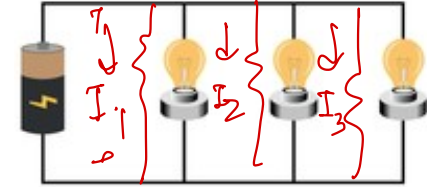
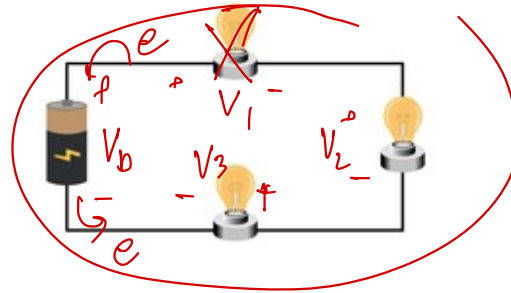


Q. Three identical 60 W, 230 V bulbs are available.

(a) You connect them all in series across a 230 V supply.

(b) You connect them all in parallel across the same 230 V supply.

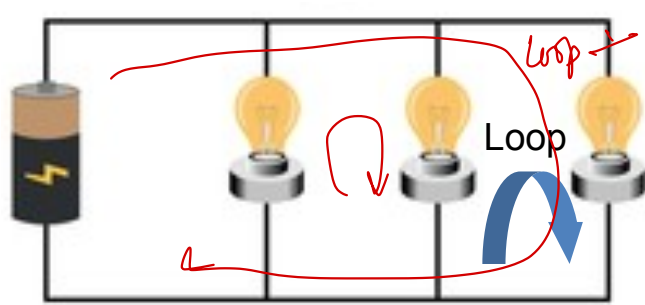
In which case will each bulb glow brighter, and why?



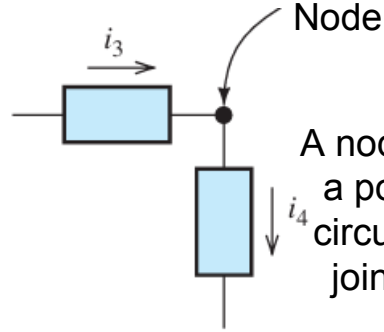
Firstly, let us answer qualitatively.

How will you go about answering it quantitatively?

Need for Laws for Circuit Analysis



Q. What constitutes a branch?



A node in an electrical circuit is a point at which two or more circuit elements/branches are joined together (similar to a junction).



In reality: Practical circuits have multiple nodes, branches and loops!

Overhead wires on a supply line



Flyover near Adalaj Junction

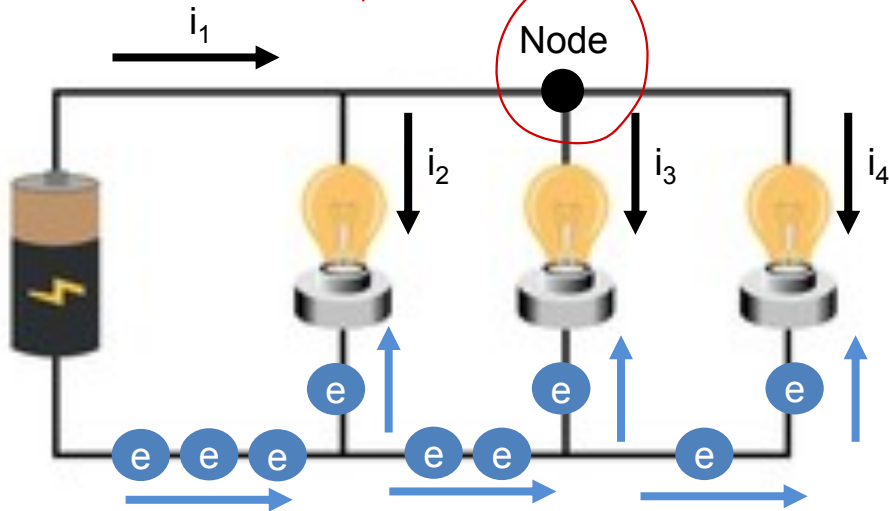


Kirchoff's Law

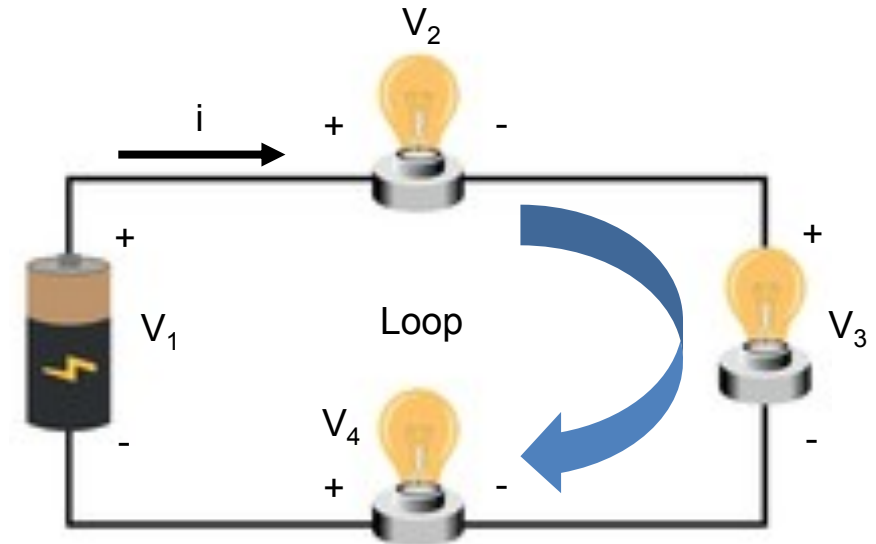
Two physical laws that are also applicable to electrical circuits:

- Conservation of Charge → **The net current entering a *node* is zero (Kirchhoff's current law).**
- Conservation of Energy → **The algebraic sum of the voltages for *any loop* equals zero (Kirchhoff's voltage law).**

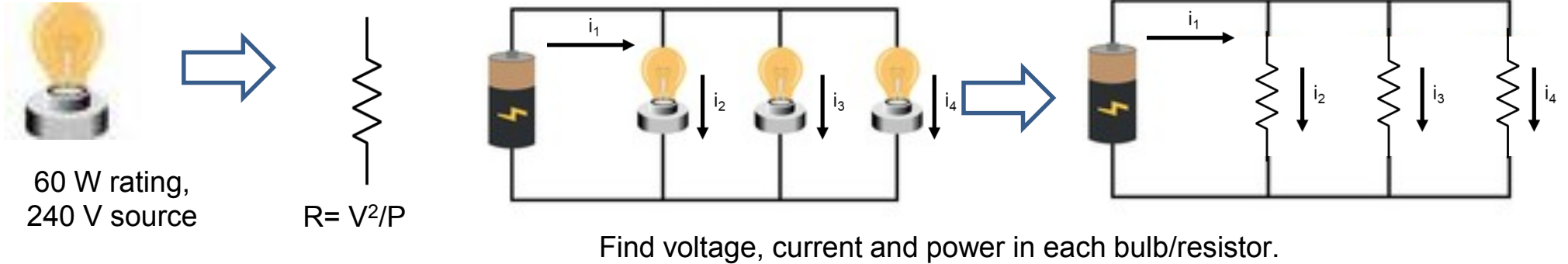
$$\begin{aligned} i_2 + i_3 + i_4 - i_1 &= 0 \\ i_1 - i_2 - i_3 - i_4 &= 0 \end{aligned}$$



$$\begin{aligned} V_1 - V_2 - V_3 - V_4 &= 0 \\ -V_1 + V_2 + V_3 + V_4 &= 0 \end{aligned}$$



Applying Kirchhoff's current law (KCL)

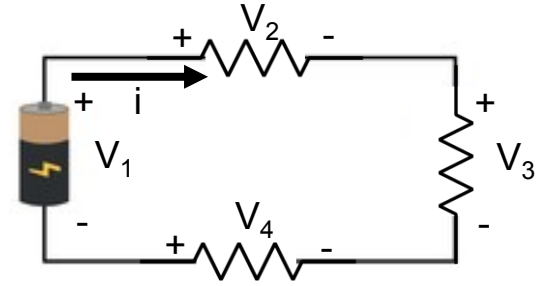
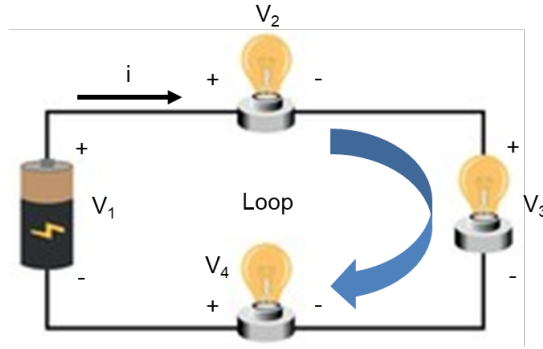
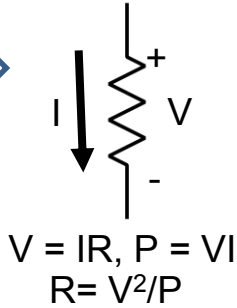
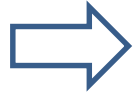


Convention: To compute the net current entering a node, we ***add the currents entering*** and ***subtract the currents leaving***.

Applying Kirchhoff's Voltage law (KVL)



60 W rating,
240 V source



Find voltage, current and power in each bulb/resistor.

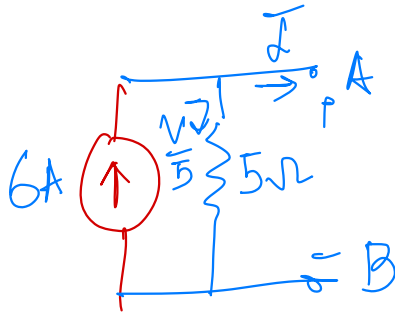
Convention: Going from the positive polarity reference to the negative reference, it carries a negative sign (voltage drop/loss of energy). Going from minus to plus, the voltage carries a positive sign (gain in energy).

Question

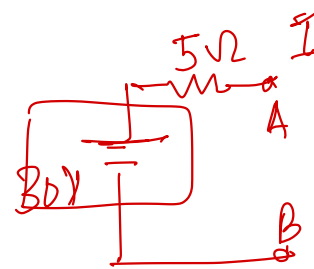
The terminal voltage and terminal current were measured on the device shown in the figure, and the values of v_t and i_t are tabulated.

- Construct a circuit model of the device inside the box.
- Using this circuit model, predict the power this device will deliver to a 10Ω resistor.

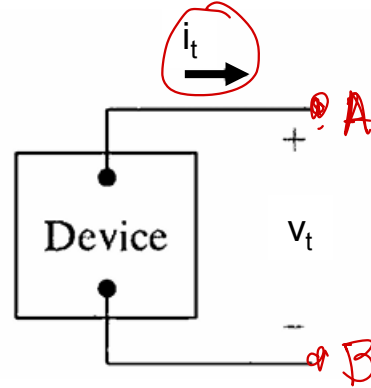
$$I = 6 - \frac{V}{5}$$



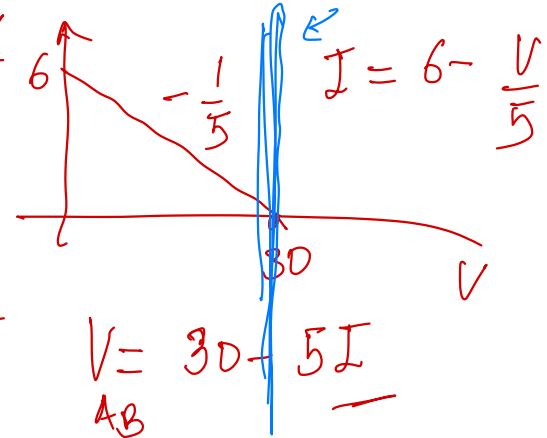
with Multimeter
Voltage measurement
Open circuit



Current measurement
Short circuit



v_t (V)	i_t (A)
30	0
15	3
0	6

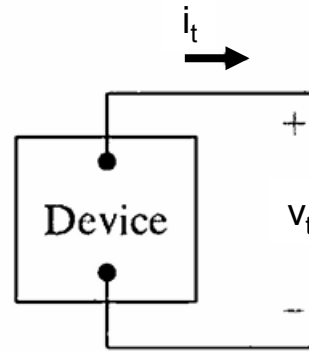


Can we construct the model using a current source?

$V = 30 - 5I \rightarrow$ representation with voltage source
or $I = 6 - \frac{V}{5} \rightarrow$ representation with current source.

I : current measurement with multimeter
(i.e. when DMM provides a short circuit path)

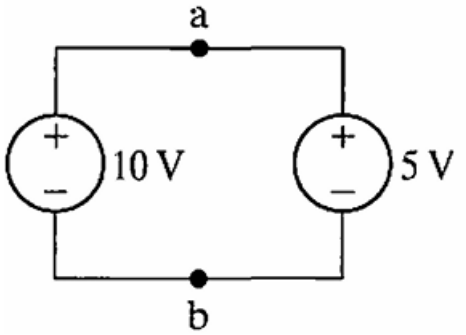
V : voltage diff. measurement with multimeter
(i.e. when DMM provides an open circuit path)



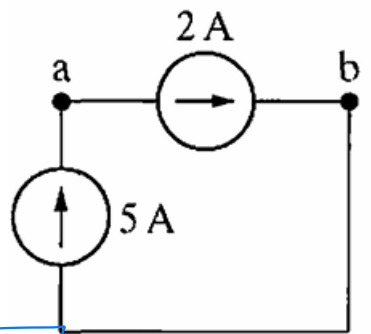
v_t (V)	i_t (A)
30	0
15	3
0	6

Which circuits are invalid?

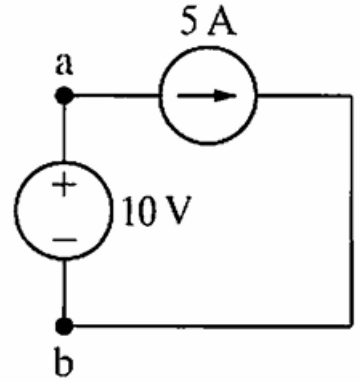
A



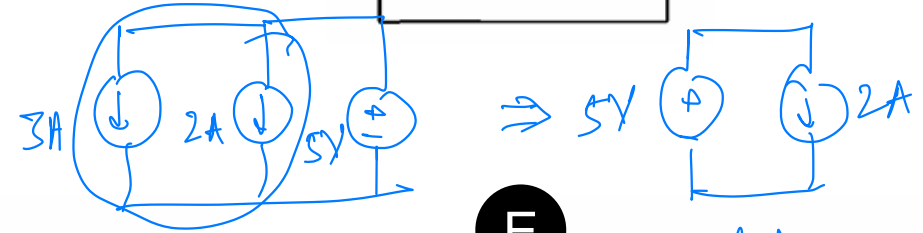
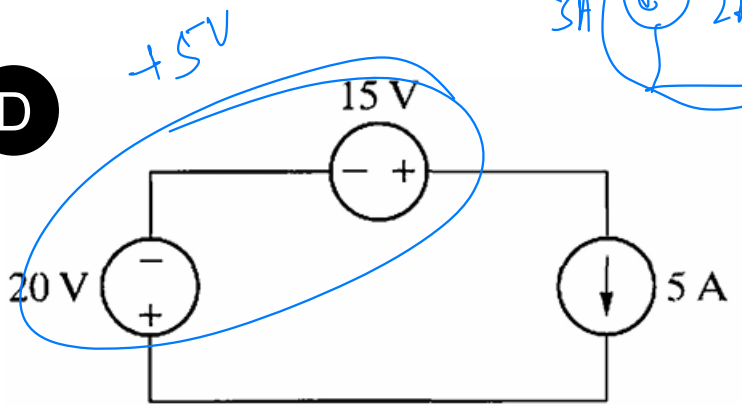
B



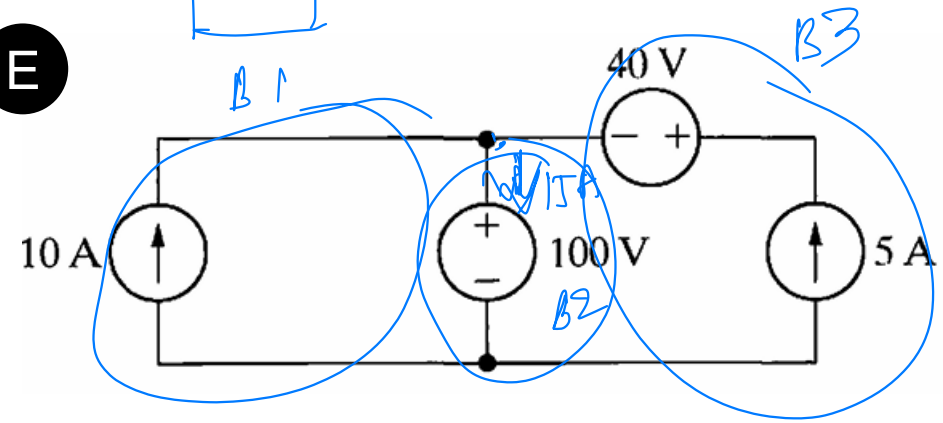
C



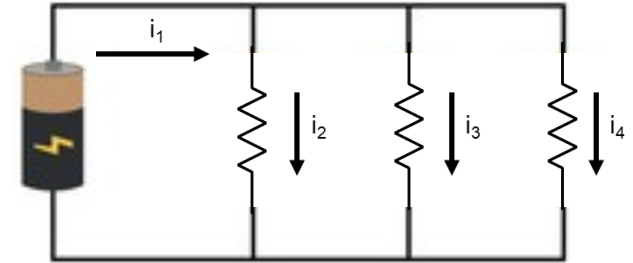
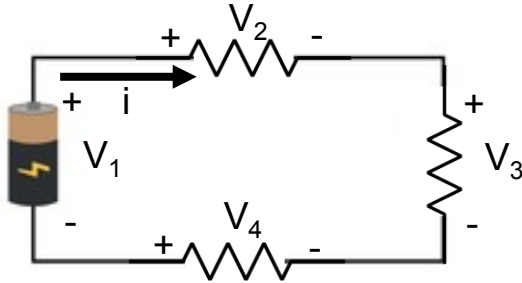
D



E



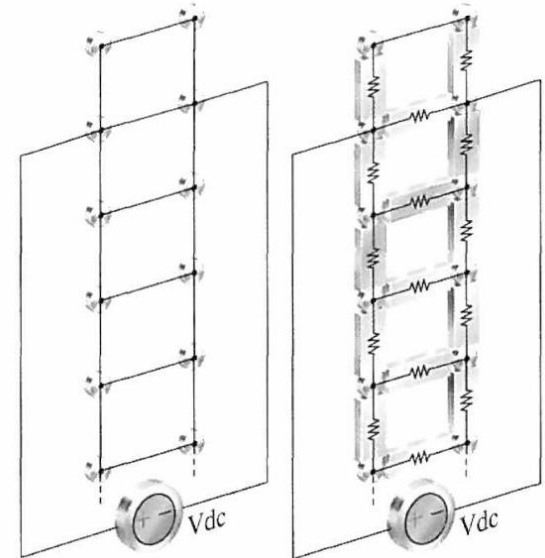
Equivalent Resistances in Series/Parallel Networks



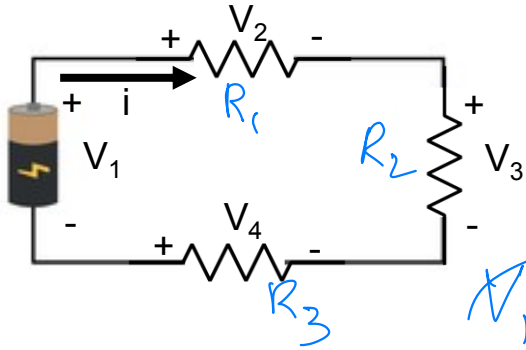
Practical Example

A rear window deforester

- The grid conductors in one such grid of a rear window deforester can be modeled with resistors, as shown in the figure.



Equivalent Resistances in Series/Parallel Networks



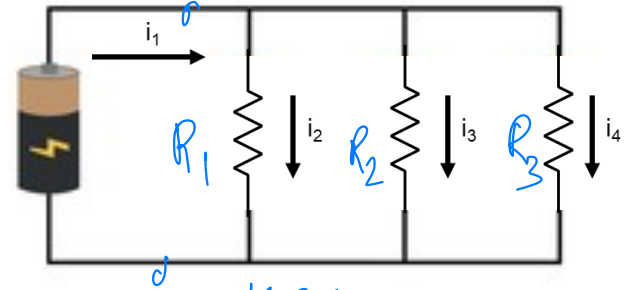
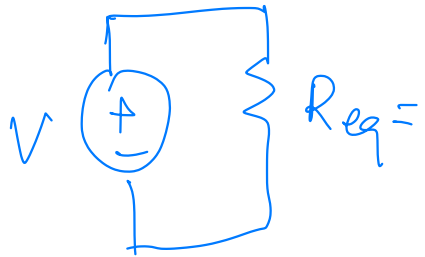
KVL

$$V_1 = V_2 + V_3 + V_4$$

$$= IR_1 + IR_2 + IR_3$$

$$= I(R_1 + R_2 + R_3)$$

$$R_{eq} = \frac{V}{I} = R_1 + R_2 + R_3$$



KCL

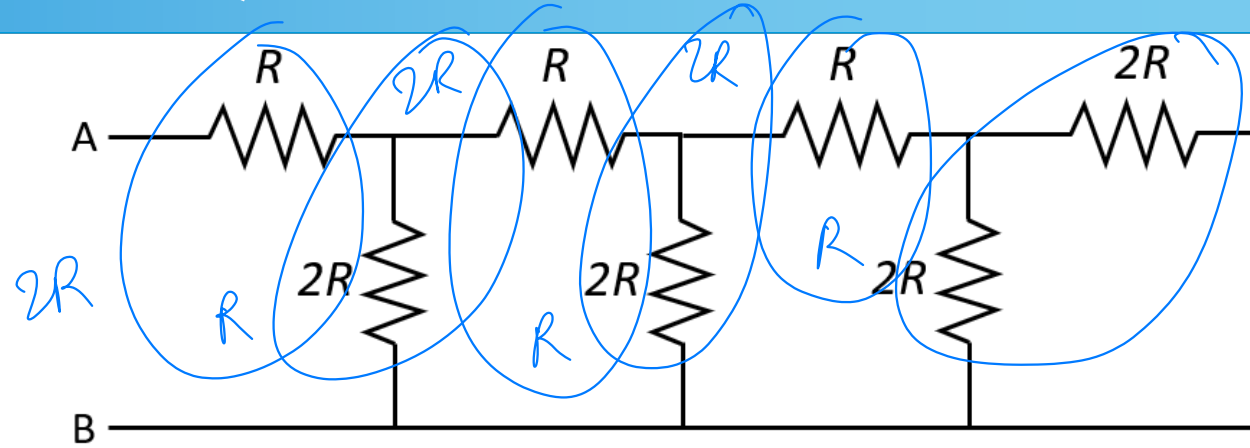
$$I_1 = I_2 + I_3 + I_4$$

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Practice Questions

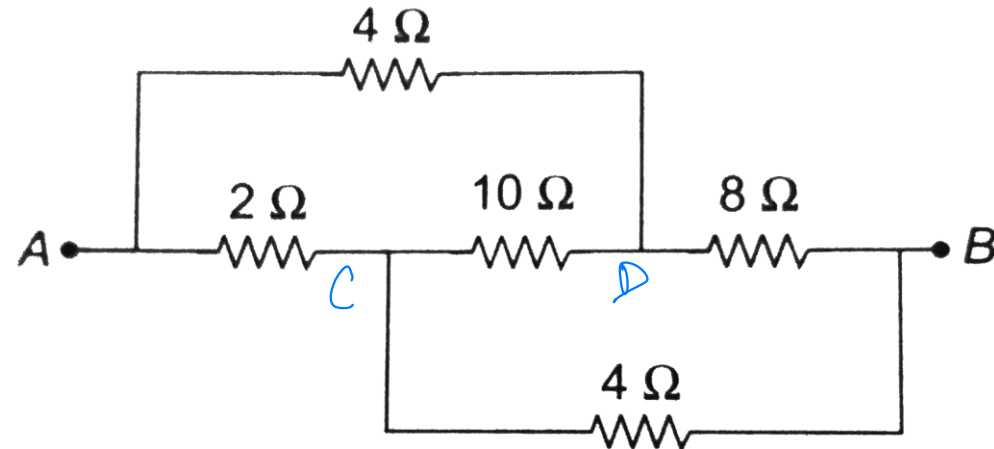
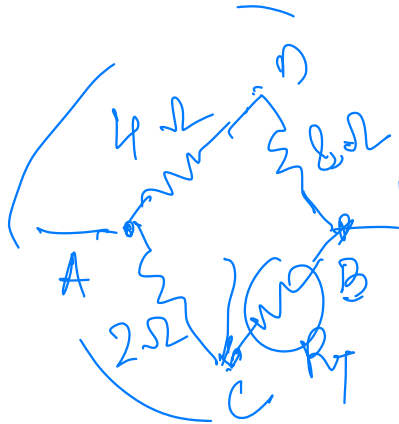
Calculate R_{AB} in terms of R .



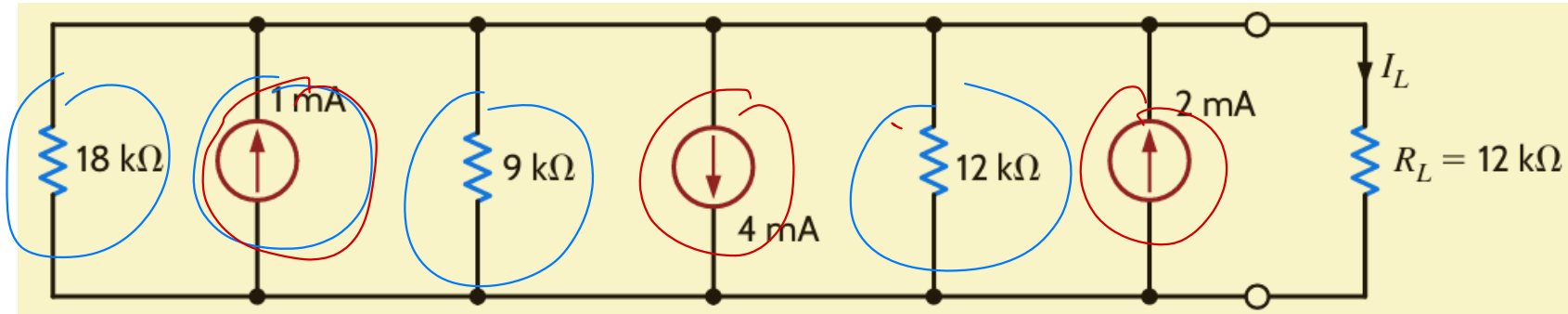
$R=2R$
ladder
DAC

Calculate R_{AB} .

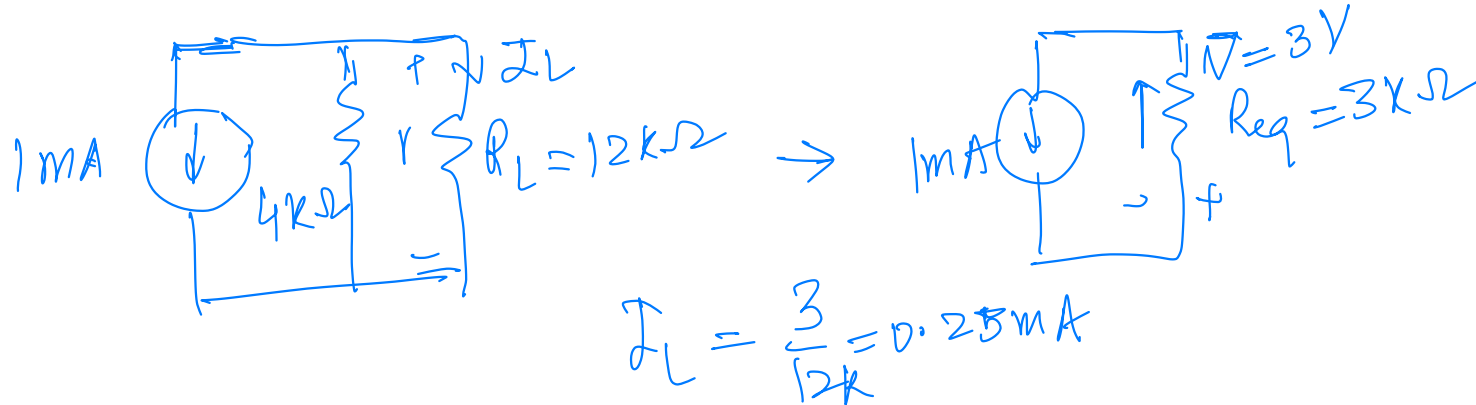
$R(T)$
thermistor
 $36.5^\circ C$



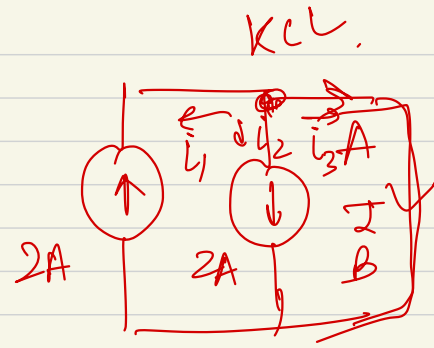
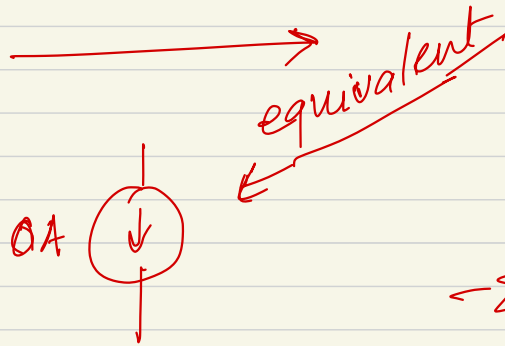
Resistive Networks with multiple sources



Q. Find the current in the 12-k Ω load resistor [Hint: Build an equivalent circuit].



Why can we sum the currents from the current sources when they are connected to the same nodes?



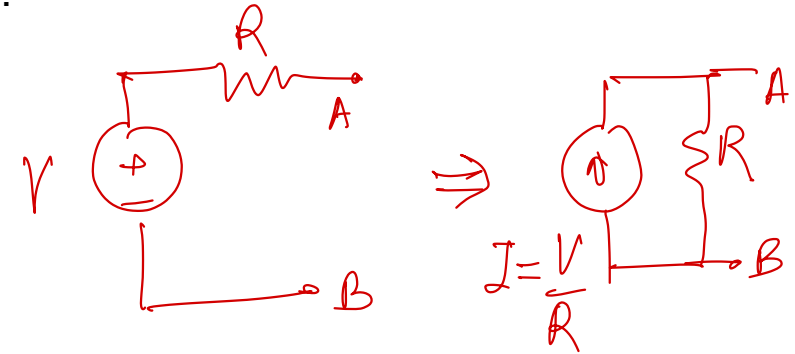
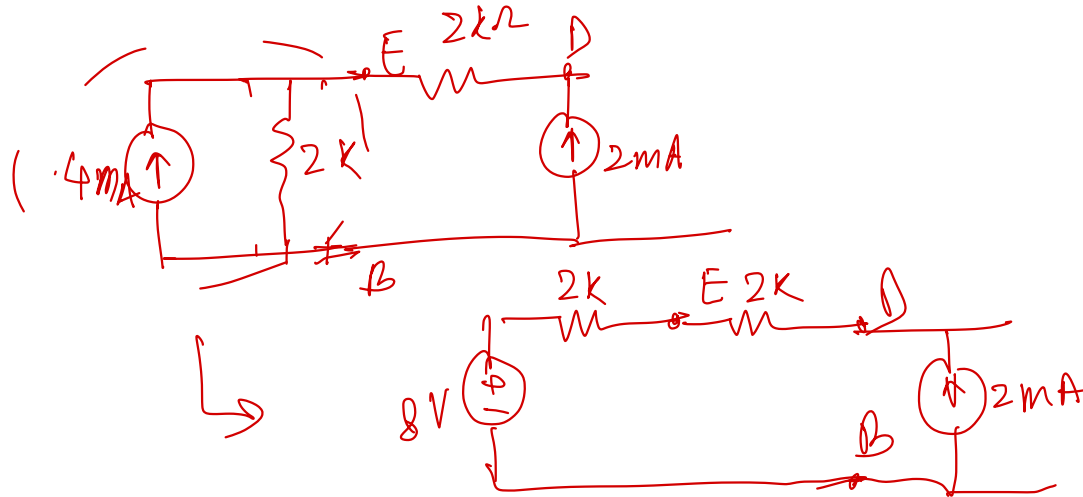
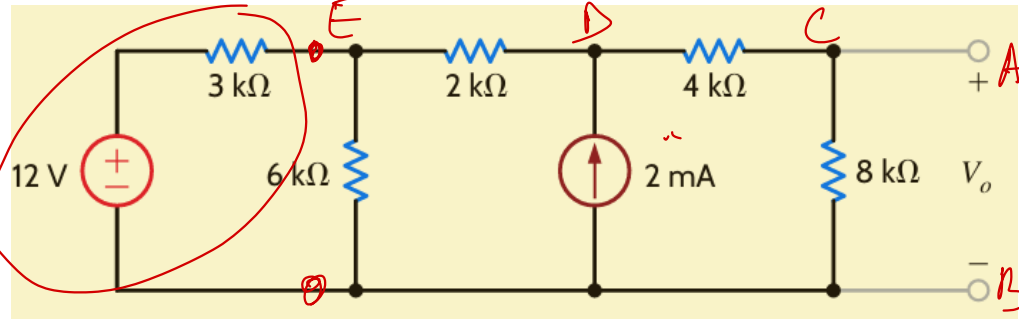
$$i_1 + i_2 + i_3 = 0$$

$$-2A + 2A + i_3 = 0$$

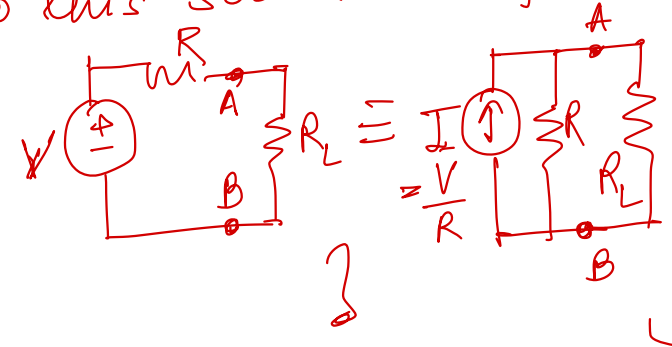
$$i_3 = 0$$

Repeated Application of Source Transformation

Q. Find V_o using the repeated application of source transformation.



Q. if A & B are connected to other elements, can I still do this source transformation?



Practice Question (if time permits)

Write KCL and KVL equations for the shown circuit that is commonly encountered in the analysis and design of transistor amplifiers.

- A. KCL at nodes a, b, c and d.
- B. KVL in loops 1, 2, 3 and 4.

